3 Propan-2-ol, $(CH_3)_2CHOH$, is sometimes added to fuel to help it burn.

Fig. 3.1 shows some reactions of propan-2-ol.

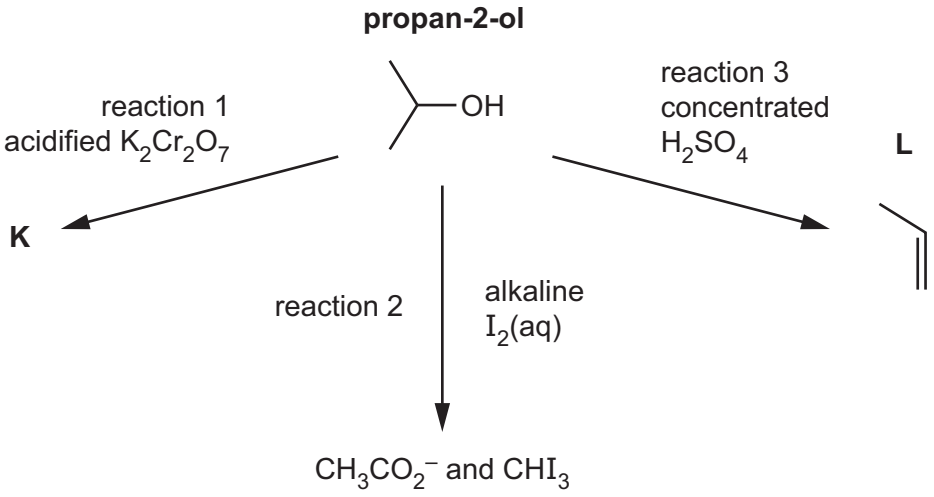


Fig. 3.1

(a) (i) Draw the structure of organic compound **K**.

[1]

(ii) State an observation you would make in reaction 2.

..... [1]

6 Three bottles of colourless liquids labelled **F**, **G** and **H** contain separate pure samples of the compounds ethanal, propanal or propanone but **not** necessarily in that order.

(a) State the functional group in ethanal, propanal and propanone.

..... [1]

(b) State a reagent and the relevant observation that confirm that **F**, **G** and **H** have the same functional group.

reagent .....

observation .....
[2]

(c) Separate samples of **F**, **G** and **H** are each tested with Tollens' reagent and with alkaline I_2(aq). The observations are shown in Table 6.1.

Table 6.1

	Tollens' reagent	alkaline I_2(aq)
F	no observable change	pale yellow precipitate
G	silver mirror	pale yellow precipitate
H	silver mirror	no precipitate

(i) Use Table 6.1 to name the organic compounds in bottles **F**, **G** and **H**.

F = ...

G = ...

H = ...
[2]

(ii) Identify the yellow precipitate produced when alkaline I_2(aq) is added to separate samples of **F** and **G**.

..... [1]

4 Fig. 4.1 shows compounds **J** to **M**, each of which contains four carbon atoms.

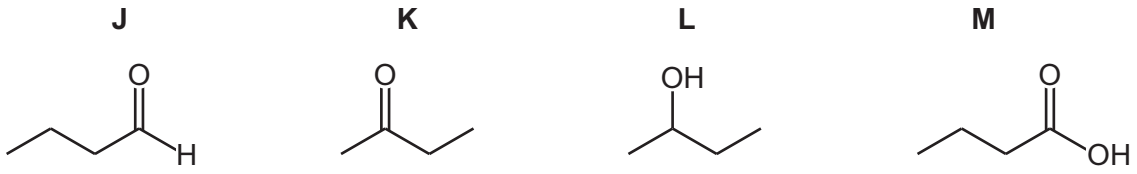


Fig. 4.1

(a) Table 4.1 gives details of tests on **J** to **M**. In each test, only **two** compounds give a positive result.

Complete Table 4.1.

Table 4.1

reagent	observation of positive result	compounds that give a positive result
acidified $K_2Cr_2O_7$(aq)		**J** and **L**
alkaline I_2(aq)	yellow precipitate	
	orange precipitate	**J** and **K**
Na(s)		

[5]

(b) **K** reacts with HCN in the presence of a KCN catalyst, forming **N**.

(i) Complete Fig. 4.2 to show the mechanism for this reaction.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.2

[4]

(ii) Identify the class of compound of which **N** is a member.

..... [1]

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2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(a) (i) Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

(ii) State the essential condition for chlorine to react with C_2H_6 at room temperature.

..... [1]

(iii) Write numbers in the boxes to balance the equation showing the reaction of propanone with NaC*l*O.

CH_3COCH_3 + ☐ NaC*l*O → ☐ CHCl_3 + ☐ CH_3COONa + ☐ NaOH

[1]

4 Propanone, CH_3COCH_3, is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

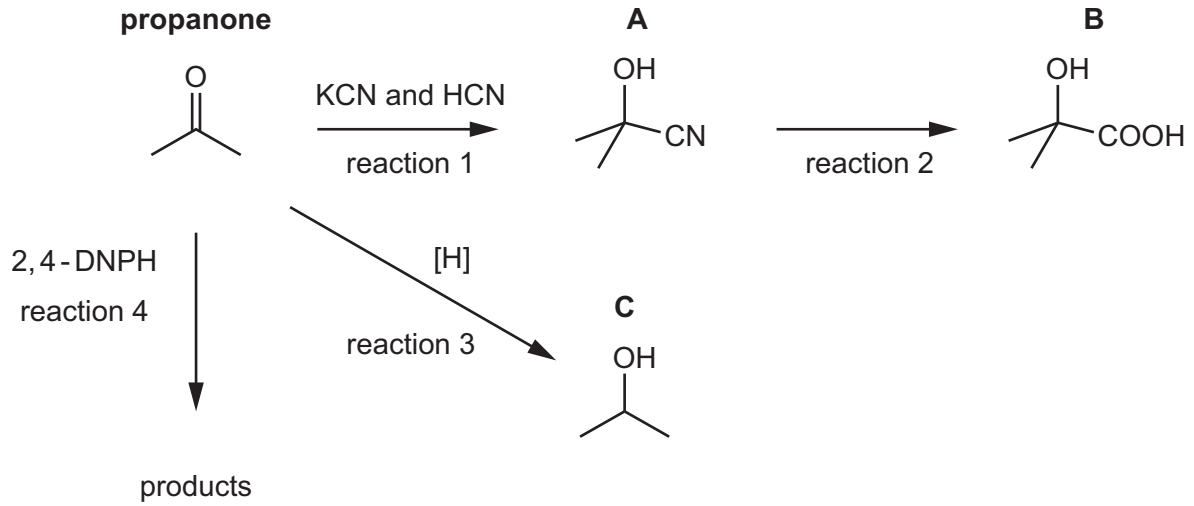


Fig. 4.1

(a) Reaction 1 is a nucleophilic addition reaction.

(i) Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

(c) Reaction 3 is a reduction reaction.

(i) Construct an equation to represent reaction 3.

Use [H] to represent one atom of hydrogen from the reducing agent.

..... [1]

(d) State what is observed in reaction 4.

..... [1]

(e) Explain why Fehling's reagent does **not** react with propanone.

.....

..... [1]

4 Bromoalkanes are used widely in industry, although there is increasing concern about their environmental impact.

Fig. 4.1 shows a reaction scheme involving 1,2-dibromoethane.

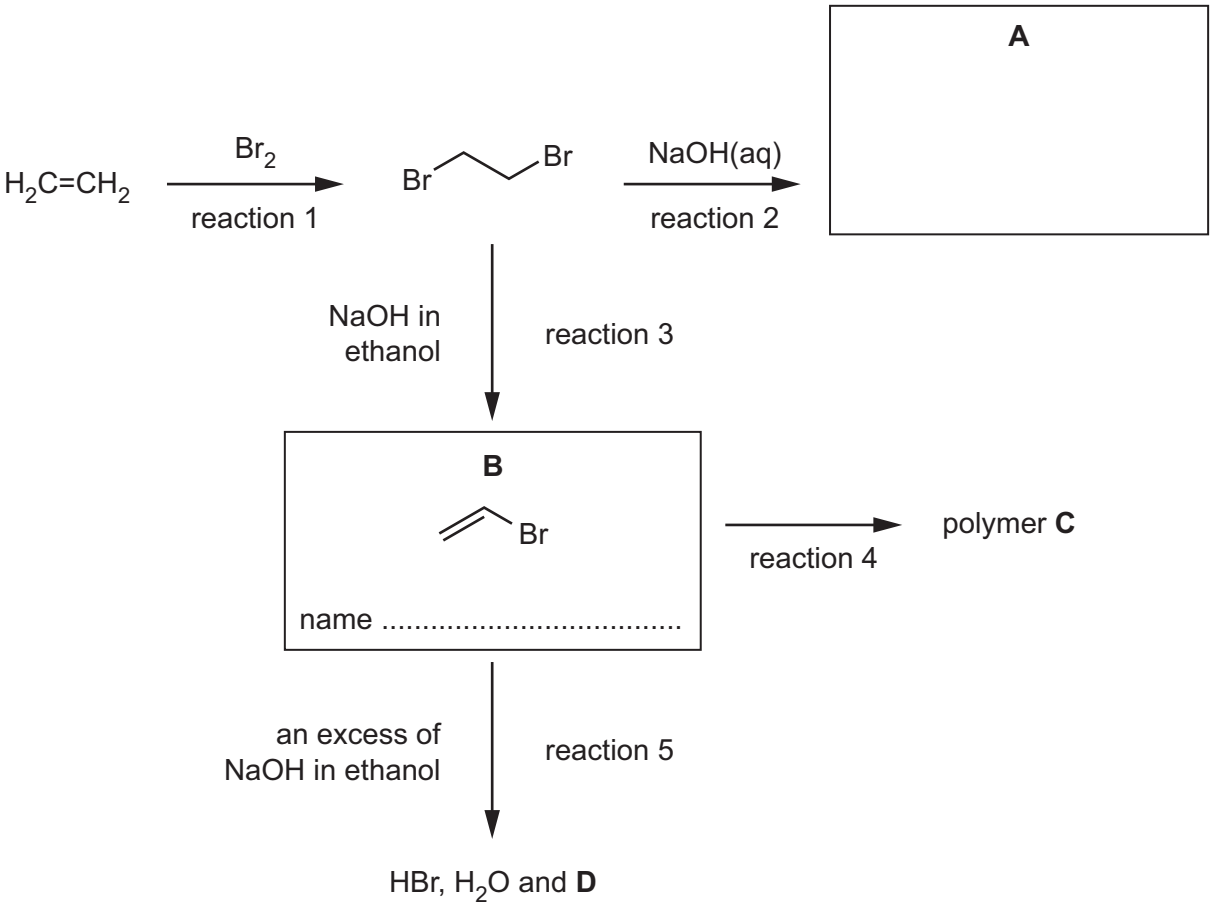


Fig. 4.1

(a) Complete Fig. 4.2 to show the mechanism for the formation of 1,2-dibromoethane in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



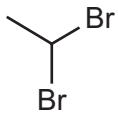
Fig. 4.2

[3]

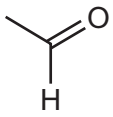
(d) Compound **E** is the only isomer of 1,2-dibromoethane.

Alkaline hydrolysis of **E** gives compound **F**.

E



F



(i) Identify the type of isomerism shown by **E** and 1,2-dibromoethane.

..... [1]

(iii) Complete Table 4.1 to state what is observed when **F** reacts with the reagents listed.

Table 4.1

reagent	observation with **F**
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	
Tollens' reagent	
alkaline I_2(aq)	

[3]

BLANK PAGE

2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(a) (i) Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

(ii) State the essential condition for chlorine to react with C_2H_6 at room temperature.

..... [1]

(iii) Write numbers in the boxes to balance the equation showing the reaction of propanone with NaC*l*O.

CH_3COCH_3 + ☐ NaC*l*O → ☐ CHCl_3 + ☐ CH_3COONa + ☐ NaOH

[1]

4 Propanone, CH_3COCH_3, is an important organic reagent. Fig. 4.1 shows some reactions of propanone and its derivatives.

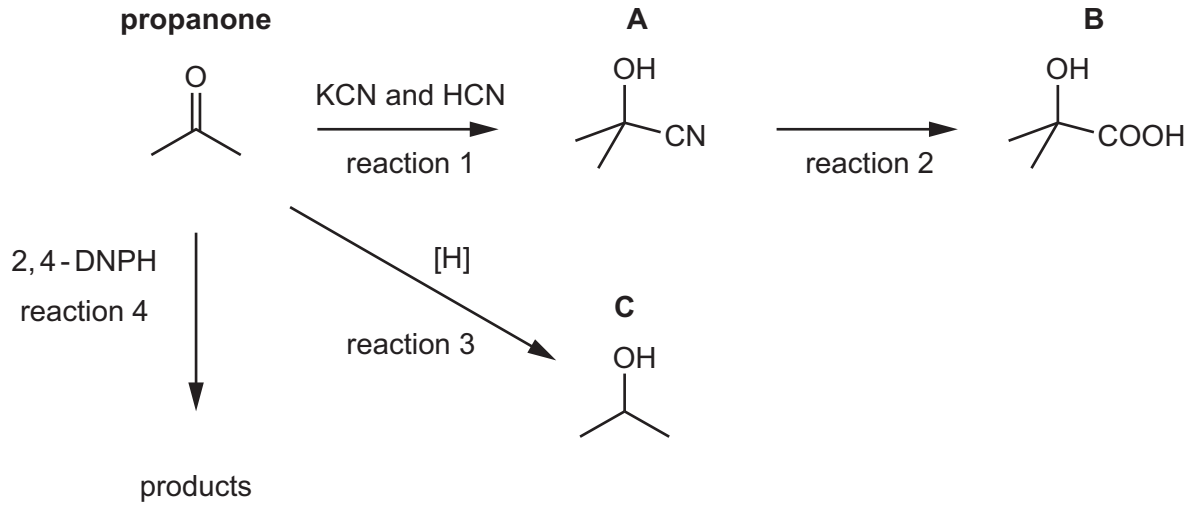


Fig. 4.1

(a) Reaction 1 is a nucleophilic addition reaction.

(i) Complete Fig. 4.2 to show the mechanism for the formation of **A** from propanone.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



Fig. 4.2

[3]

(c) Reaction 3 is a reduction reaction.

(i) Construct an equation to represent reaction 3.

Use [H] to represent one atom of hydrogen from the reducing agent.

..... [1]

(d) State what is observed in reaction 4.

..... [1]

(e) Explain why Fehling's reagent does **not** react with propanone.

.....

..... [1]

5 Compound **W** has molecular formula $C_4H_{10}O$. It contains only **one** functional group.

(a) Table 5.1 shows the two peaks with the greatest *m*/*e* values in the mass spectrum of **W**.

Table 5.1

m/*e*	relative abundance
74	50
75	x

(i) Calculate the relative abundance, x, of the peak at *m*/*e* = 75 using the information from Table 5.1.

x = ... [1]

(ii) The mass spectrum of **W** also shows peaks at *m*/*e* = 29 and *m*/*e* = 59.

Suggest the molecular formulae of these fragments.

m/*e* = 29 .....

m/*e* = 59 .....

[2]

(b) A sample of **W**, $C_4H_{10}O$, is heated under reflux with an excess of acidified $K_2Cr_2O_7$ until there is no further reaction. Only **one** organic product, **X**, is present in the mixture at the end of the reaction.

Fig. 5.1 shows the infrared spectrum of **W**.

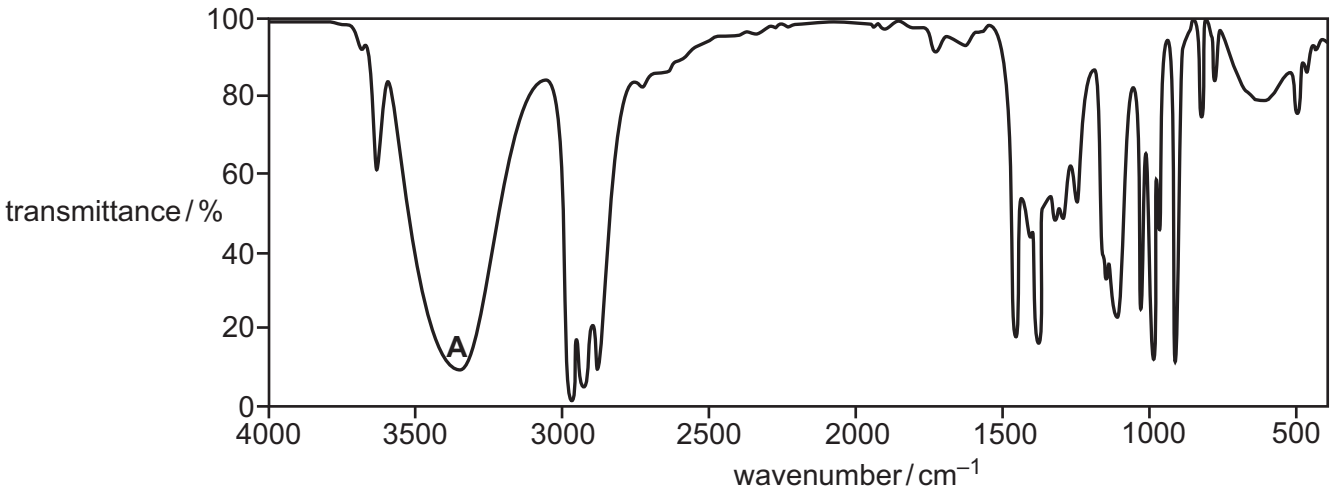


Fig. 5.1

Fig. 5.2 shows the infrared spectrum of **X**.

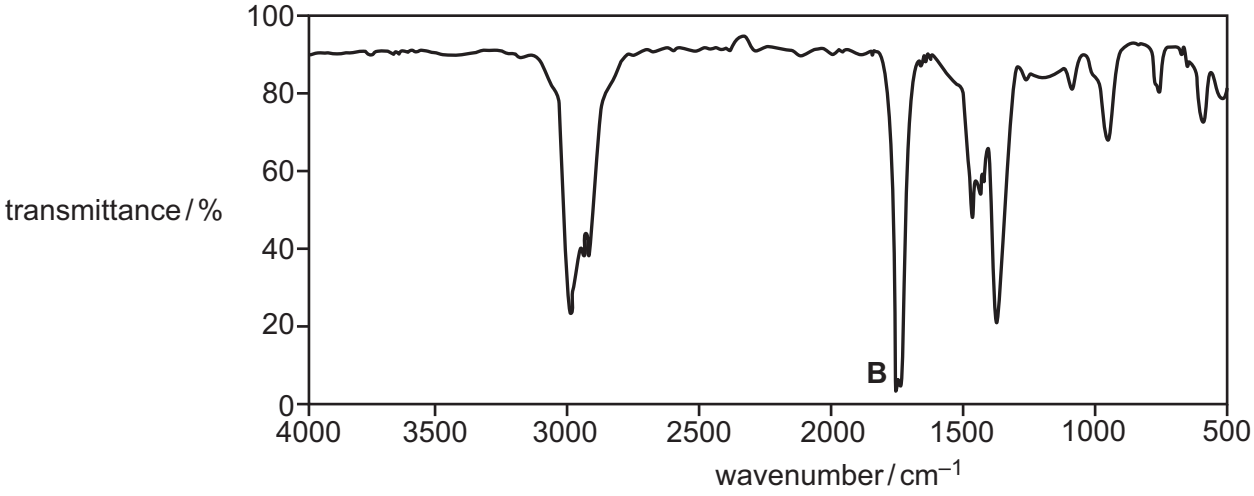


Fig. 5.2

Table 5.2

bond	functional groups containing the bond	characteristic infrared absorption range (in wavenumbers) / cm^{-1}
C–O	hydroxy, ester	1040–1300
C=C	aromatic compound, alkene	1500–1680
C=O	amide carbonyl, carboxyl ester	1640–1690 1670–1740 1710–1750
C≡N	nitrile	2200–2250
C–H	alkane	2850–2950
N–H	amine, amide	3300–3500
O–H	carboxyl hydroxy	2500–3000 3200–3650

(i) Absorption **A** is shown in Fig. 5.1.
Absorption **B** is shown in Fig. 5.2.

Complete Table 5.3 using the information given in Fig. 5.1, Fig. 5.2 and Table 5.2.

Table 5.3

absorption	bond	functional group containing the bond
A		
B		

[1]

(ii) Use the information in **(a)** and **(b)(i)** to draw the structure of **X** in the box in Fig. 5.3.

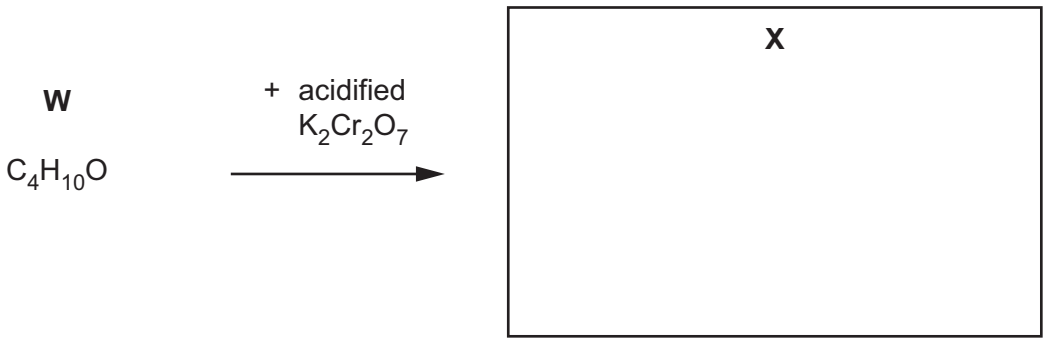


Fig. 5.3

[1]

6 **(a)** **V** shows stereoisomerism.

V

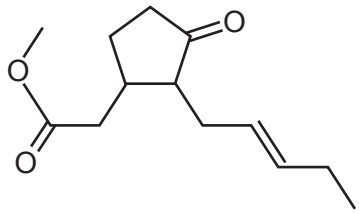


Fig. 6.1

(i) Explain what is meant by stereoisomerism.

.....

.....

..... [1]

(c) Both functional groups in one molecule of **Y** react with an inorganic reagent to form one molecule of **Q** and one molecule of methanol, CH_3OH, as shown in Fig. 6.3.

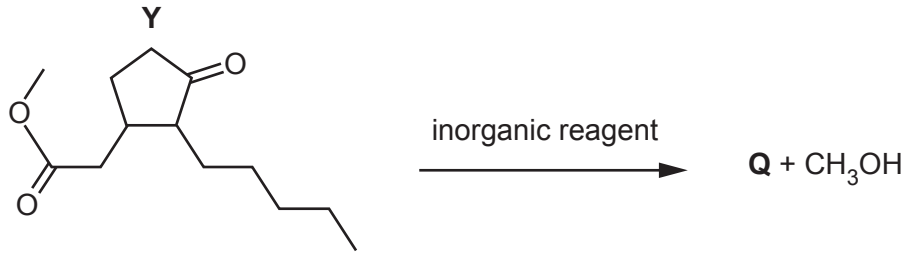


Fig. 6.3

(i) Part of the mass spectrum for **Q** is shown in Fig. 6.4. Only peaks with *m*/*e* greater than 198 are shown.

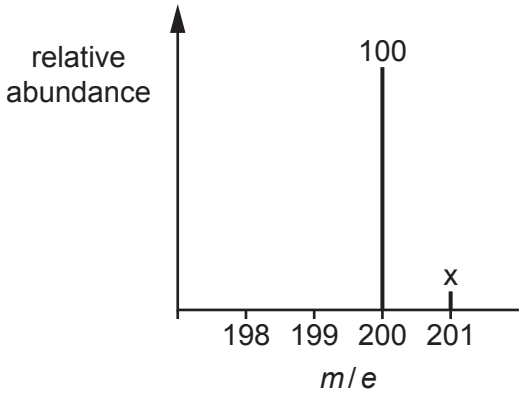


Fig. 6.4

Calculate the relative abundance, x, of the peak at *m*/*e* = 201.

Show your working.

x = ... [2]

(ii) **Q** contains **only** hydroxyl functional groups.

Complete Table 6.1 to show the observations that occur when 2,4-dinitrophenylhydrazine (2,4-DNPH reagent) is added to separate samples of **Y** and **Q**.

Table 6.1

	observation on addition of 2,4-DNPH reagent
Y	
Q	

[1]

1 Bismuth is an element in Group 15 of the Periodic Table.

(a) Bismuth has metallic bonding.

Draw a labelled diagram to show the metallic bonding in bismuth.

[1]

(d) $NaBiO_3$ is an oxidising agent with similar properties to $KMnO_4$.

Fig. 1.1 shows an example of the use of $NaBiO_3$ as an oxidising agent.

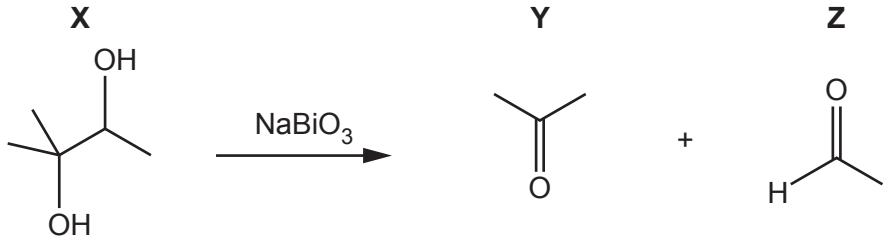


Fig. 1.1

(i) Explain the term oxidising agent.

.....

..... [1]

(ii) Compound **X** forms when methylbut-2-ene reacts with $KMnO_4$.

State the essential conditions for this reaction.

..... [1]

(iii) Complete Table 1.1 to show what is observed when compounds **Y** and **Z** react separately with the named reagents.

Table 1.1

reagent	observation with **Y**	observation with **Z**
$Na_2CO_3(aq)$	no reaction	
alkaline $I_2(aq)$		
2,4-dinitrophenylhydrazine (2,4-DNPH)		
Tollens' reagent		

[4]

(iv) Construct an equation for the reaction of **Z** with $NaBH_4$.

Use [H] to represent an atom of hydrogen from the reducing agent.

..... [1]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

(e) The structure of glucose, $C_6H_{12}O_6$, is shown in Fig. 3.2.

glucose

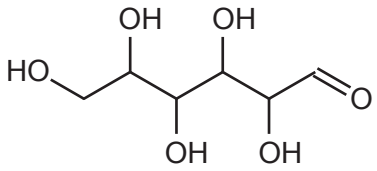


Fig. 3.2

(i) Complete Table 3.1 to identify the number of primary, secondary and tertiary alcohol groups present in the structure shown in Fig. 3.2.

Table 3.1

type of alcohol group	primary	secondary	tertiary
number of groups			

[1]

(ii) Separate samples of aqueous glucose are tested with the reagents shown in Table 3.2.

Complete Table 3.2 with the observation for each reaction.

Write “no reaction” if applicable.

Table 3.2

reagent and conditions	observation with glucose
acidified $KMnO_4(aq)$ and warm	
Fehling’s reagent and warm	
alkaline $I_2(aq)$ and warm	

[3]

4 Lactic acid, $CH_3CH(OH)COOH$, and pyruvic acid, $CH_3COCOOH$, both contain two functional groups.

lactic acid **pyruvic acid**

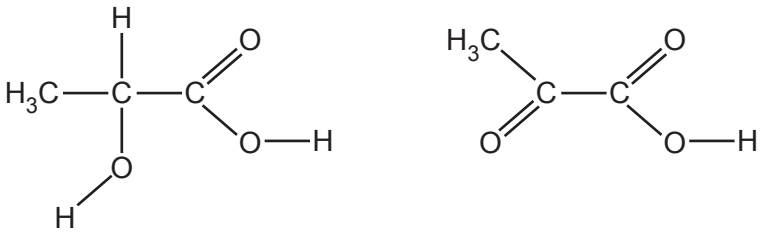


Fig. 4.1

(a) (i) Explain why lactic acid exists as optical isomers.

.....

.....

..... [1]

(b) Two possible syntheses of pyruvic acid are shown in Fig. 4.3 and Fig. 4.4.

Each synthesis has a total of three steps.

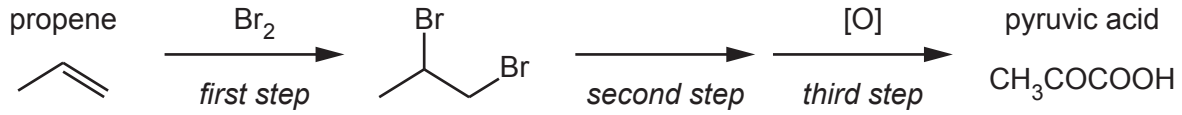


Fig. 4.3

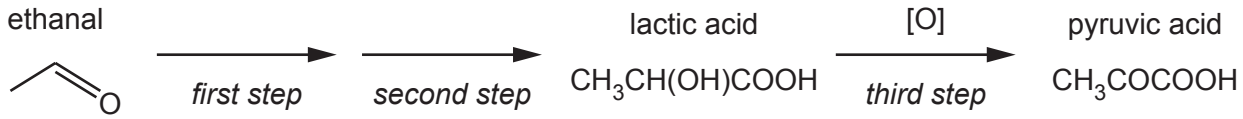


Fig. 4.4

(i) Complete the diagram in Fig. 4.5 to show the mechanism for the reaction of propene with Br_2.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.



Fig. 4.5

[3]

(iii) Complete Table 4.1 to give details of the reagents and conditions used in each of the two syntheses shown in Fig. 4.3 and Fig. 4.4.

Table 4.1

		synthesis from propene (shown in Fig. 4.3)	synthesis from ethanal (shown in Fig. 4.4)
reagents and conditions used	*first step*	Br_2	
	second step		
	third step		

[4]

3 Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

(a) Define oxidising agent.

.....

..... [1]

(e) The structure of glucose, $C_6H_{12}O_6$, is shown in Fig. 3.2.

glucose

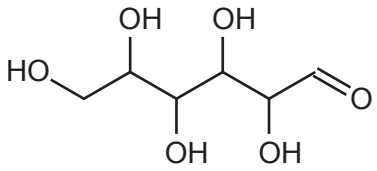


Fig. 3.2

(i) Complete Table 3.1 to identify the number of primary, secondary and tertiary alcohol groups present in the structure shown in Fig. 3.2.

Table 3.1

type of alcohol group	primary	secondary	tertiary
number of groups			

[1]

(ii) Separate samples of aqueous glucose are tested with the reagents shown in Table 3.2.

Complete Table 3.2 with the observation for each reaction.

Write “no reaction” if applicable.

Table 3.2

reagent and conditions	observation with glucose
acidified $KMnO_4(aq)$ and warm	
Fehling’s reagent and warm	
alkaline $I_2(aq)$ and warm	

[3]

5 **Y** is formed from **X** in a single-step reaction, as shown in Fig. 5.1.

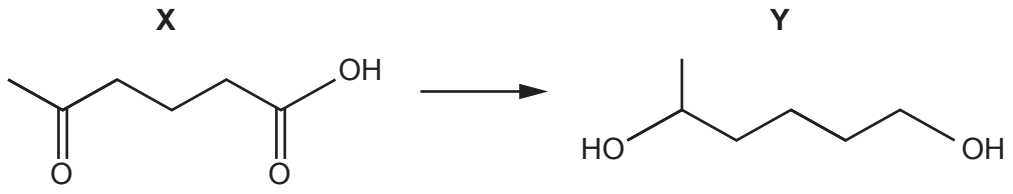


Fig. 5.1

(a) Deduce the empirical formula of **Y**.

..... [1]

(b) The formation of **Y** from **X** requires the addition of a suitable reducing agent.

(i) Construct an equation using molecular formulae and [H] for the reaction in Fig. 5.1. Use [H] to represent one atom of hydrogen from the reducing agent.

..... [1]

(ii) Identify a suitable non-gaseous reducing agent for the formation of **Y** from **X**.

..... [1]

(d) Complete Table 5.2 with the expected observations that occur when the reagents shown are added to separate solutions of **X** and **Y**. Do **not** refer to temperature changes in your answer.

Table 5.2

reagent	observation on addition to **X**	observation on addition to **Y**
aqueous sodium carbonate		
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)		
alkaline aqueous iodine		

[3]

6 Compound **W**, CH_2=CHCN, is used to make an addition polymer which is present in carbon fibres.

(a) Draw **one** repeat unit of the addition polymer of **W**.

[1]

(b) CH_3CHO is used in a two-step synthetic route to form **W**, as shown in Fig. 6.1.
In step 1, CH_3CHO is heated with HCN in the presence of KCN.

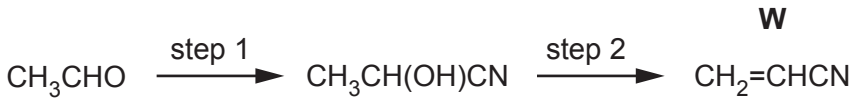


Fig. 6.1

(i) Name the mechanism for the reaction in step 1 in Fig. 6.1.

..... [1]

(ii) Complete Fig. 6.2 to show the mechanism for the reaction in step 1.
Include all products, charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

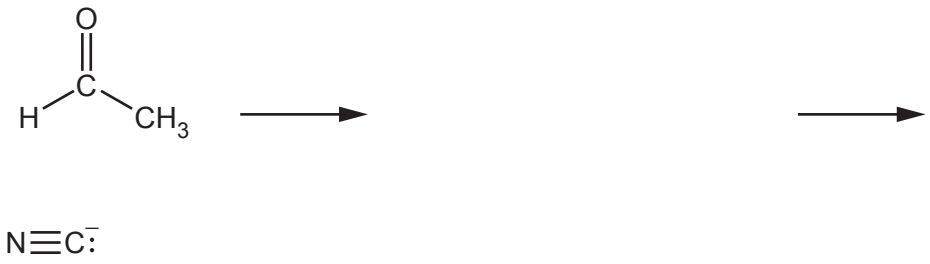


Fig. 6.2

[3]

5 **(a)** Describe structural isomerism.

.....

..... [1]

(b) **A** and **B** are structural isomers with molecular formula $C_5H_{10}O$.

They are both straight-chained molecules with only one functional group.

Table 5.1 describes observations when separate samples of **A** and **B** are added to different reagents.

Table 5.1

reagent	**A**	**B**
2,4-dinitrophenylhydrazine (2,4-DNPH reagent)	orange precipitate appears	orange precipitate appears
Tollens' reagent	silver mirror appears	no reaction
alkaline I_2(aq)	no reaction	no reaction

(i) Name the functional group present in both **A** and **B**.

..... [1]

(ii) Draw the structures of **A** and **B** in the boxes.

A

B

[2]

3 Alkenes undergo an addition reaction with a 1:1 mixture of CO and H_2 to form aldehydes.

Fig. 3.1 shows the reaction of propene with a 1:1 mixture of CO and H_2.

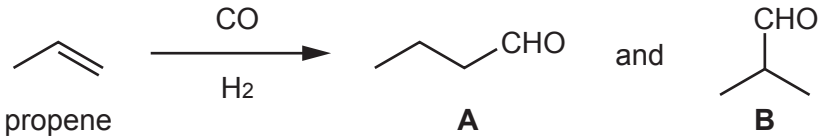


Fig. 3.1

(a) (i) Define addition reaction.

.....

..... [1]

(iii) Name **A**.

..... [1]

(b) **A** and **B** show reactions typical of aliphatic aldehydes.

(i) **A** undergoes a nucleophilic addition reaction with a mixture of HCN and KCN, forming compound **C**.

Complete the diagram to show the mechanism for this reaction.

Include charges, dipoles, lone pairs of electrons and curly arrows, as appropriate.

Draw the structure of the organic intermediate.



[4]

(ii) Table 3.1 shows information about three experiments involving **B**.

Complete Table 3.1.

Table 3.1

experiment	reagents	observation with **B**
1		solution turns from orange to green
2		a silver mirror forms on the sides of the reaction vessel
3	$Br_2(aq)$	

[3]